Vinicius Vaz Cascais

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Professional Summary

Software Engineer specializing in **cloud-native infrastructure**. At Capgemini Engineering, working with Nokia as the client, I run zero-downtime **Kubernetes deployments** with **Helm** for production systems **serving 5M+ customers**, and configure traffic routing for highly available, **multi-site distributed systems**. Strong systems foundation from 42 Lisboa (C/C++, Linux, process management, multithreading) and hands-on experience with Docker, Nginx, Terraform, and CI/CD pipelines.

Professional Experience

Software Engineer - Capgemini Engineering *July 2026 - Present | Lisbon*

- Performed **version upgrades** of 5G/4G subscriber registers (UDM/HSS/HLR) in production for a network **serving 5M+ customers**, troubleshooting and resolving deployment issues in real time to keep services within carrier-grade availability requirements.
- Executed **live Kubernetes rolling updates** of telecom production using Helm and kubectl, delivering **zero-downtime deployments** at a cadence of ~2 production rollouts per month.
- Configured traffic routing across d**istributed systems** using Kubernetes Ingress and internal networking stacks, enabling **high availability** and **multi-site redundancy** for critical telecom services.
- Integrated **internal LLM tools** into the troubleshooting workflow, cutting incident troubleshooting time **by roughly 80%.**

Full Stack Developer (Intern) - Ebinar *July 2025 - October 2025 | Lisbon*

- Designed and implemented an Excel-to-MySQL automation tool to replace manual imports, **increasing efficiency by ~25%** and reducing data inconsistencies, while keeping track and displaying filterable logs for the user.

Education

42 Lisboa - Lisbon *Ongoing*

Software Engineering Bachelor

Focus on C/C++, Object-Oriented Programming (OOP), system fundamentals and low-level memory management

IEFP - Lisbon *July 2025*

Network and Systems Administration – Technical Specialist Course

Learned programming fundamentals (Python, PHP, and introductory OOP)

Built and maintained Windows and Linux servers, configured networks with Cisco Packet Tracer

Technical Skills

Programming Languages: C, C++, Bash, PHP

Cloud & Containers: Kubernetes, Red Hat OpenShift, Docker, Helm, Terraform, AWS, Google Cloud, VMware

Systems & Infrastructure: Linux (RHEL, Arch, Ubuntu, Debian), Server Administration, Nginx, CI/CD Pipelines

AI & Developer Tools: Claude Code (AI-assisted development and integration), Cursor, Ollama (local LLMs), ChatGPT, Perplexity, Agentic AI workflows, Machine Learning Fundamentals

Databases: SQL, MySQL/MariaDB

Networking: TCP/IP, FTP, UDP, DNS, Load Balancing, Kubernetes Ingress, Multi-site Routing

Tools: Git, GDB, Valgrind

Projects

Minishell – Custom Unix Shell in C

- Created a Unix-like shell in C that handles processes, pipes, redirections, environment variables, and built-in commands. Expanded it with advanced features like logical operators and wildcard support to loosely replicate Bash .

Webserver - C++ Local Webserver

- HTTP/1.1 web server from scratchwith no external libraries, using a single non-blocking poll() event loop that handles multiple ports, static files, uploads, and CGI execution (Python, Node.js) without crashing or hanging on malformed or slow clients. Built an NGINX-style configuration parser supporting per-route methods, redirects, autoindex, and body size limits, validated by an automated test suite of 120+ malformed configurations, siege stress tests, and Valgrind memory checks.

Inception - System Administration & Container Orchestration

- Designed and deployed a multi-container infrastructure using Docker and Docker Compose, virtualizing a full-stack environment with Nginx, MariaDB, and WordPress.

Languages

Portuguese (Mother Tongue), English(C1), French (A2)

Certifications

L3 Kubernetes Certification (Nokia)

Multiple cloud certifications (Azure, Nokia, AWS)